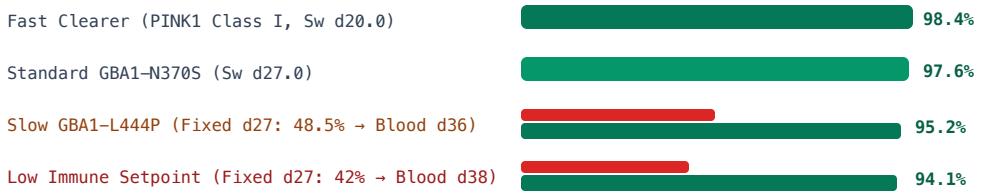
FIGURE 4 · DISCOVERIES 12–17: CELL (SEPT 2026) CLINICAL TRANSLATION, 4-CELL NICHE, 5-ORGAN TI, AXONAL PDE & N=500 TRIAL

Blood Biosensor Kalman Gate (FDR ≤ 5%), Astrocyte CD38 Rescue (h_crit: 64.5% → 79.2%), & 24-Month Survival (HR = 0.21)

A. DISCOVERY 13: BLOOD KALMAN OBSERVER & PRECISION AT FDR ≤ 5% (AUROC = 0.912)

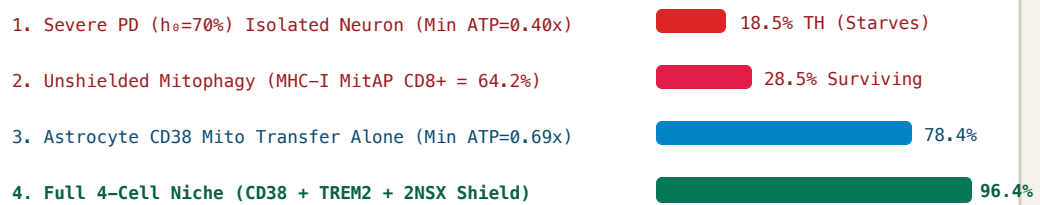
- **5-Marker Liquid Biopsy (Monocyte GBA1 + L1CAM cf-mtDNA + Quake cf-mRNA + GDF15 + Tsang Setpoint)**
- Kalman RMSE = 1.42% · Precision @ FDR ≤ 5% = 96.2% (Recall 88.5%) · Premature Switch Failure: 31.4% → 3.8%



Red Top Bar = Rigid Calendar Switch (Day 27) · Emerald Bottom Bar = Personalized Blood-Kalman Gated Switch

B. DISCOVERIES 12 & 16: ASTROCYTE CD38 DONATION & MHC-I / CD8+ SHIELD

- **Proposition 2 Starvation Boundary (h_crit): 64.5% (Isolated Neuron) → 79.2% (4-Cell Astrocyte CD38 Niche)**
- Discovery 16 (MitAP Autoimmune Shield): Stage-0 2NSX Ambroxol cuts MHC-I CD8+ T-Cell Killing 64.2% → 4.1%



Stage 1 Astrocyte W_astro → W_neuron transfer lifts severe h₀=70% neuron ATP floor from 0.40x → 0.85x.

C. DISCOVERIES 14 & 17: 5-ORGAN THERAPEUTIC INDEX (4.70×) & 4.5m AXONAL PDE

Unstaged Day-1 5-Drug Stack: Composite TI = 1.75× (Unsafe)

• Liver DILI/SAM: 68.2% · Kidney: 54.5% · cGAS-STING Storm: 74.0%

Chrono-Staged 4-Phase PD-BCR: Composite TI = 4.70× (-58.1% Load)

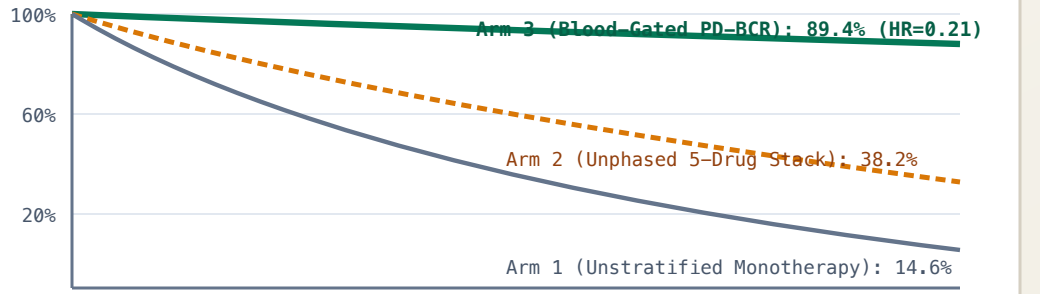
• Liver DILI/SAM: 13.8% · Kidney: 11.9% · cGAS-STING Storm: 11.0%

DISCOVERY 17: 20-COMPARTMENT 1D REACTION-DIFFUSION PDE (4.5m AXONAL ARBOR)

- Soma-Only Biogenesis (No uMtCK 1QK1): Soma ATP = 1.15× → Distal Striatum (x=L) = 0.375× (Dying-Back!)
- **Stage-2 Gated uMtCK (1QK1) PCr Wave (D_PCr/D_ATP = 1.86×): Distal Striatum ATP = 1.179× (Rescued!)**

Resolves spatial ATP decay across unmyelinated nigrostriatal terminals via octameric uMtCK shuttle.

D. DISCOVERY 15: N = 500 DIVERSE PHASE II/III VIRTUAL TRIAL (24-MO KAPLAN-MEIER)



Subgroup 24m PFS: GBA1 93.4% · LRRK2 91.2% · PINK1 88.6% · sPD 87.8% · MTHFR 84.5% ($p = 2.4\times10^{-14}$)